

Supplementary Material

Automated Specification Inference via Heuristic Static Analysis for Program Comprehension and Test Generation

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This document provides supplementary material for the main article, including detailed annotation type descriptions, loop invariant heuristic data, a worked inference example, LLM prompt templates, and an approach comparison table.

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1 Appendix A: Generated Annotation Types

Unlike traditional JML tools that embed specifications in Javadoc comments, JML-Inferrer generates Java annotations that are compiled into bytecode and queryable at runtime via reflection. Table 1 lists the annotation types, all defined in the `com.jml.inferrer.annotations` package with runtime retention.

Specifications use JML-like syntax [1, 3] with the following constructs:

- `\result` – The method’s return value
- `\old(expr)` – Value of `expr` at method entry
- `\nothing` – Empty frame (no modifications)
- `this.field` – Instance field reference
- `arr[*]` – All array elements
- `(\forall int k; range; prop)` – Universal quantification

Table 1: Annotation types generated by JML-Inferrer

Annotation	Target	Description
<i>Contract Specifications</i>		
@Requires	Method	Precondition (repeatable)
@Ensures	Method	Postcondition (repeatable)
@Signals	Method	Exceptional postcondition
@LoopInvariant	Method	Loop invariant (repeatable)
@Invariant	Class	Class invariant (repeatable)
@Assignable	Method	Frame condition
<i>Method Classification</i>		
@Pure	Method	No side effects, deterministic
@Observer	Method	Reads but doesn't modify state
@Mutator	Method	Modifies object state
@ThreadSafe	Method/Class	Thread-safe implementation
<i>Nullability</i>		
@Nullable	Field/Param	May be null
@NonNull	Field/Param	Never null
<i>Additional Properties</i>		
@Immutable	Field/Class	Immutable value
@Complexity	Method	Time/space complexity
@MustCall	Method	Required cleanup calls

2 Appendix B: Loop Invariant Inference Heuristics

Loop invariant inference is a key challenge in automated verification [2, 4]. Our tool employs a combination of heuristics that have proven effective in practice.

Heuristic 1: Bound-Based Invariants. For loops with a clear iteration variable and bound:

```
for (int i = 0; i < n; i++) { ... }
```

We infer: $0 \leq i \wedge i \leq n$.

Heuristic 2: Accumulator Patterns. For loops that accumulate values:

```
int sum = 0;
for (int i = 0; i < n; i++) {
    sum += arr[i];
}
```

We track the accumulator variable and infer bounds based on symbolic unrolling.

Heuristic 3: Monotonicity Detection. We detect whether loop variables are monotonically increasing or decreasing by analyzing the update expression:

- $i++$ or $i += k$ (for positive k): monotonically increasing
- $i--$ or $i -= k$ (for positive k): monotonically decreasing

Heuristic 4: Symbolic Unrolling. For loops where pattern-based heuristics fail, we perform bounded symbolic unrolling (default: 3 iterations) to detect emergent patterns. If a consistent relationship is observed, it is generalized to an invariant candidate.

Heuristic 5: While-Loop Counter Detection. For `while` loops, which lack explicit initialization and update expressions, the tool scans the loop body for counter variables—identified by increment/decrement operations (`i++`, `i--`, `i += n`) and compound assignments. Detected counters are then combined with bounds extracted from the loop condition (e.g., `while (i < n)` yields $i \geq 0$ and $i < n$). The same accumulator and variable-relationship analyses used for `for` loops are applied to the body. Common iterator idioms such as `while (iter.hasNext())` are also recognized.

Heuristic 6: Enhanced For-Each Analysis. For enhanced `for-each` loops, the tool infers element membership invariants (`elem` is an element of the iterable) and collection-size bounds (processed elements $\leq$ collection size). Variable-relationship analysis (e.g., max/min tracking) is applied to the loop body, producing richer invariants than element membership alone.

Fallback Behavior. When all heuristics fail, the tool produces a conservative postcondition that does not constrain the loop’s effect. We record this as a “weak specification” for analysis.

Table 2 reports the success rate of each heuristic on the 68 loops encountered across 48 looping methods in our Commons Lang evaluation dataset.

Table 2: Loop invariant inference success rates		
Heuristic	Loops	Success Rate
Bound-based (for)	32	93.8%
Accumulator pattern	15	80.0%
Monotonicity	13	84.6%
Symbolic unrolling	8	62.5%
While-loop counter	—	—
Enhanced for-each	—	—
Overall	68	85.3%
Fallback (weak spec)	10	—

3 Appendix C: Worked Example — End-to-End Inference

This appendix presents a complete example showing how JML-Inferer analyzes a method and derives its specification.

3.1 Input Method

Consider the following utility method that validates and returns an array element:

```
public static String getElement(String[] arr, int index) {
    if (arr == null) {
        throw new IllegalArgumentException("Array cannot be null");
    }
    if (index < 0 || index >= arr.length) {
```

```

        throw new IndexOutOfBoundsException("Invalid index");
    }
    return arr[index];
}

```

Listing 1: Input method before specification inference

3.2 Precondition Inference (WP-Inspired)

The tool identifies guard-and-throw patterns that correspond to preconditions:

1. **First if-statement:** The pattern `if (arr == null) throw ...` is recognized as a null-check guard, yielding precondition `arr != null`.
2. **Second if-statement:** The pattern `if (index < 0 || index >= arr.length) throw ...` is recognized as a range-check guard, yielding preconditions `index >= 0` and `index < arr.length`.

These heuristic inferences correspond to what a formal WP analysis would derive—the weakest precondition for the return statement `return arr[index]` with postcondition `true` is exactly `arr != null ∧ 0 ≤ index < arr.length`.

Inferred Preconditions: `arr != null, index >= 0, index < arr.length`

3.3 Postcondition Inference (SP-Inspired)

The tool analyzes the return statement to derive postconditions:

1. The return expression `arr[index]` is an array access on input parameters, yielding `\result == arr[index]`.
2. Since no fields are modified, the method is classified as `@Pure`.

Inferred Postcondition: `\result == arr[index]`

3.4 Output: Annotated Method

```

@Requires("arr != null")
@Requires("index >= 0")
@Requires("index < arr.length")
@Ensures("\result == arr[index]")
@Assignable("\nothing")
@Pure
@Complexity(time = "O(1)", space = "O(1)")
public static String getElement(String[] arr, int index) {
    if (arr == null) {
        throw new IllegalArgumentException("Array cannot be null");
    }
    if (index < 0 || index >= arr.length) {
        throw new IndexOutOfBoundsException("Invalid index");
    }
    return arr[index];
}

```

Listing 2: Output method with inferred JML specifications

The tool also infers that the method is `@Pure` (no side effects) and has constant time complexity.

4 Appendix D: LLM Prompt Templates

The following prompts were used for LLM-based test generation in each experimental phase. Prompts were kept minimal, differing only in the inclusion of specifications or source code.

4.1 Phase 1 Prompt (Signature Only)

Write unit tests for the following function signature under typical development constraints. Provide the tests as a developer would normally write them, without any formal specification guidance:

[method signature]

4.2 Phase 2 Prompt (Signature + Guidance)

Given the following function signature, write a comprehensive set of unit tests. Ensure that the tests cover normal behavior, edge cases, and potential failure scenarios:

[method signature]

4.3 Phase 3 Prompt (Signature + Specification)

Given the following function signature and formal specification, write a comprehensive set of unit tests. Ensure that the tests cover normal behavior, edge cases, and potential failure scenarios. The specification uses JML notation where @requires specifies preconditions and @ensures specifies postconditions:

[method signature with JML specification]

4.4 Phase 4 Prompt (Source Code)

Given the following function implementation, write a comprehensive set of unit tests. Ensure that the tests cover normal behavior, edge cases, and potential failure scenarios:

[full method source code]

5 Appendix E: Approach Comparison

Table 3 summarizes the key differences between our approach and related specification inference methods.

Table 3: Comparison of specification inference approaches						
Approach	Input	Technique	Lang.	Prec.	Rec.	Cov.
Our tool	Source	Heuristic static	Java	94.2%	89.3%	99.0%
Jdoctor	Javadoc	NLP	Java	92%	83%	45.2%
PreInfer	Execution	Symbolic	C#	N/R	N/R	N/R
Daikon	Traces	Dynamic	Multi	varies	varies	varies
LLM-based	Source	Neural	Multi	81.3%	88.7%	100%

Key observations:

- Our approach achieves the highest precision among automated methods, due to its conservative heuristic pattern matching grounded in common code idioms.
- LLM-based inference achieves higher recall but lower precision, and lacks determinism.
- Jdoctor achieves high precision but is limited by documentation availability.
- Dynamic approaches like Daikon require execution traces and may miss corner cases not exercised during testing.

References

- [1] Patrice Chalin, Joseph R. Kiniry, Gary T. Leavens, and Erik Poll. Beyond assertions: Advanced specification and verification with JML and ESC/Java2. *Formal Methods for Components and Objects*, 4111: 342–363, 2006. doi: 10.1007/11804192_16.
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